

MODEL NAME: Titan LTE
Version: MP

Revision History and Current Revision :

Revision	Description Of Changes	Date (M-D-Y)	Phase
A00	Initial schematic		P1
A01	P2 schematic		P2

Page Description

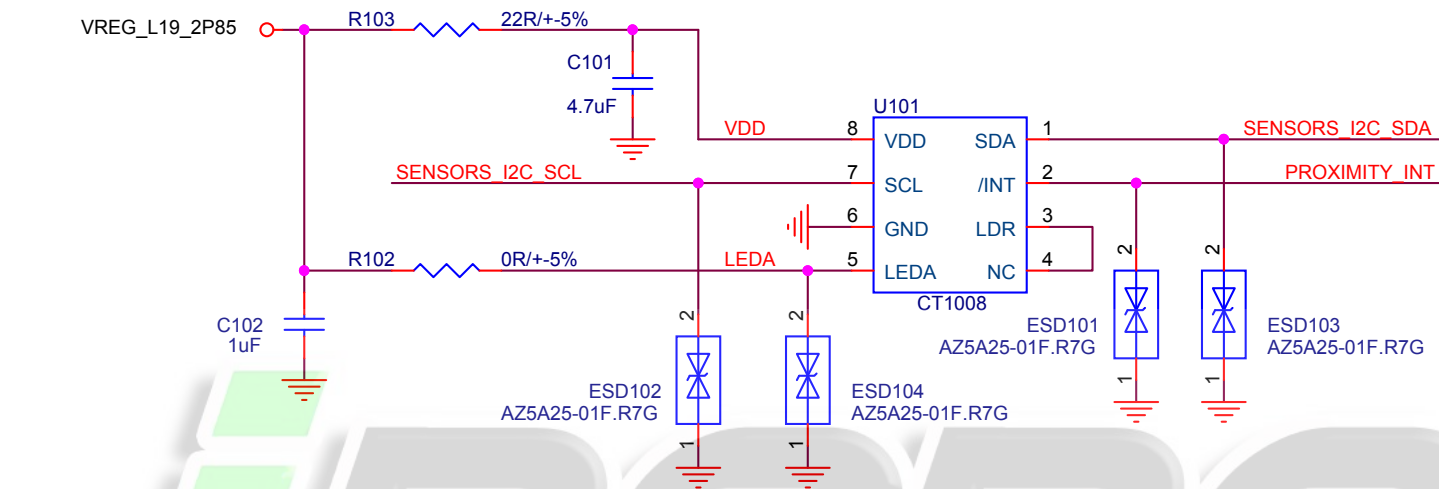
Application Pages Summary

Modem Pages Summary

01: TITLE
02: BLOCK DIAGRAM
03: GPIO/I2C LIST
04: MSM8x26_CONTROL/USB/SDC
05: MSM8x26_MIPI/RF
06: MSM8x26_GPIO
07: MSM8x26_POWER-1
08: MSM8x26_POWER-2
09: MSM8x26_GND
10: PM8926_CHARGING
11: PM8926_CONTROL/CLK/GPIO
12: PM8926_SMPS
13: PM8926_LDO
14: eMMC/LPDDR2
15: WCD9302
16: SPK/RCV/MIC/HEADSET
17: SIM/SD/VIBRATOR/FLASH LED
18: DISPLAY/TOUCH
19: CAMERA/FLASH LED DRIVER
20: KEY/TP
21: SENSOR/LED

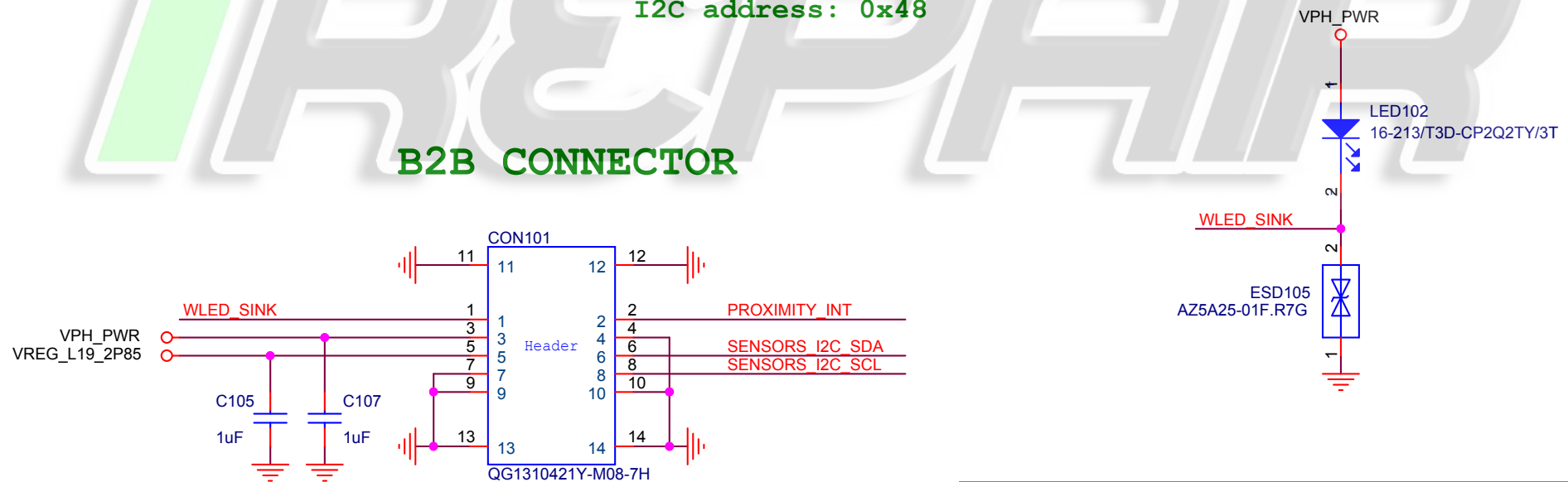
22: PRI ANT SWITCH
23: HCPA 2GQB/B1/B2/B4/B5/B8
24: Duplexer B1/B4/B3/B2/B5/B8
25: QFE1101
26: WTR1625 (L) RFIO/INTF
27: WTR1625 (L) PWR DISTRIBUTIO
28: TDD B41 TXRX
29: B12/B17 TRX
30: B28(B20) PATRX
31: B7_Tx/Rx
32: DRX FEM
33: DRX SAW
34: GNSS FRONT END
35: WCN3620

ALS + Proximity sensor



I2C address: 0x48

Indicator LED



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Title

P+ALS SENSOR/LED

Size

Document Number

TLE

Rev

P2

Date:

Thursday, September 11, 2014

Sheet

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of

1

GPIO	Net Name	Wakeup
GPIO_0		
GPIO_1		V
GPIO_2		
GPIO_3		
GPIO_4	BL_INT_N	V
GPIO_5	BL_EN	V
GPIO_6	SENSORS_I2C_SDA	
GPIO_7	SENSORS_I2C_SCL	
GPIO_8	DBG_UART_TX	
GPIO_9	DBG_UART_RX	V
GPIO_10	LCD_IOVDD_EN	
GPIO_11	LCM_ID0	
GPIO_12	LCD_BIAS_P_EN	
GPIO_13	LCD_BIAS_N_EN	V
GPIO_14	BL_I2C_SDA	
GPIO_15	BL_I2C_SCL	
GPIO_16	TP_RST_N	
GPIO_17	TP_INT_N	V
GPIO_18	TP_I2C_SDA	
GPIO_19	TP_I2C_SCL	
GPIO_20		
GPIO_21		V
GPIO_22	FLASH_I2C_SDA	
GPIO_23	FLASH_I2C_SCL	
GPIO_24	LCD_TE	
GPIO_25	DISP_RESET_N	
GPIO_26	CAM_MCLK0	
GPIO_27	CAM_MCLK1	V
GPIO_28	RCAM_AVDD_EN	
GPIO_29	CAM_I2C_SDA	V
GPIO_30	CAM_I2C_SCL	
GPIO_31	FLASH_TORCH	V
GPIO_32	FLASH_STROBE	
GPIO_33	RCAM_DOVDD_EN	V
GPIO_34		
GPIO_35	FCAM_STANDBY_N	V

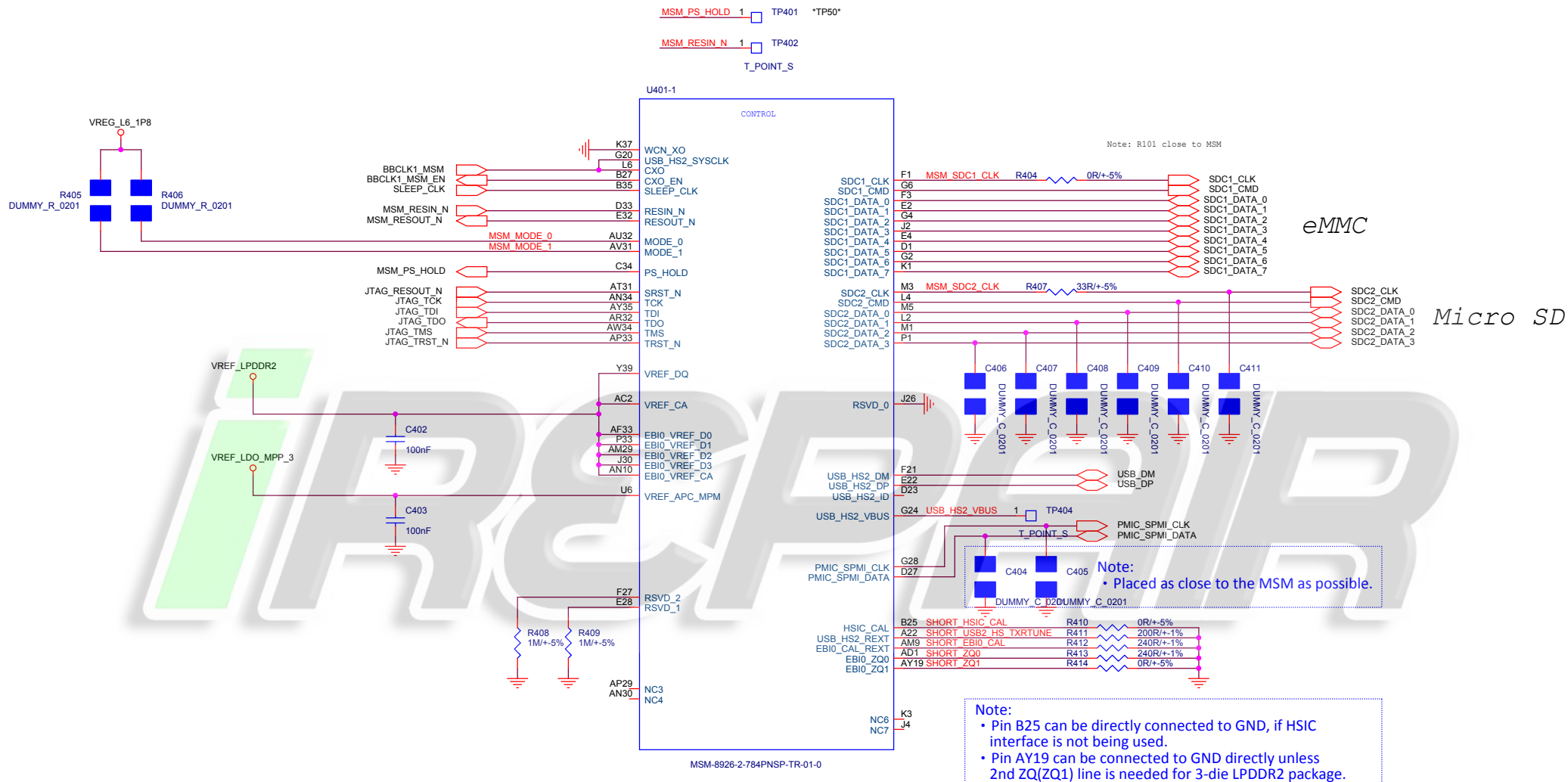
GPIO	Net Name	Wakeup
GPIO_36	FCAM_DOVDD_EN	
GPIO_37	RCAM_RST_N	V
GPIO_38	SD_CARD_DET_N	V
GPIO_39	WCSS_BT_SSBI	V
GPIO_40	WCSS_WLAN_DATA2	
GPIO_41	WCSS_WLAN_DATA1	V
GPIO_42	WCSS_WLAN_DATA0	
GPIO_43	WCSS_WLAN_SET	
GPIO_44	WCSS_WLAN_CLK	
GPIO_45	WCSS_FM_SSBI	
GPIO_46	WCSS_FM_SDI	V
GPIO_47	WCSS_BT_CTL	
GPIO_48	WCSS_BT_DAT_STB	V
GPIO_49	G-SENSOR_INT2	V
GPIO_50	TOP_SPK_AMP_EN	V
GPIO_51	BOT_SPK_AMP_EN	V
GPIO_52	SPK_RCV_SEL	V
GPIO_53	UIM2_DATA	
GPIO_54	UIM2_CLK	V
GPIO_55	UIM2_RST	
GPIO_56	UIM2_DETECT	
GPIO_57	UIM1_DATA	
GPIO_58	UIM1_CLK	
GPIO_59	UIM1_RST	
GPIO_60	UIM1_DETECT	
GPIO_61	BATT_UIM_ALARM	
GPIO_62	GYRO_INT2_N	V
GPIO_63	G-SENSOR_INT1	V
GPIO_64	GYRO_INT1_N	V
GPIO_65	PROXIMITY_INT	V
GPIO_66	E-COMPASS_INT	V
GPIO_67	LCM_ID1	V
GPIO_68	CODEC_INT	V
GPIO_69	TP_AVDD_EN	V
GPIO_70	SLIMBUS_CLK	
GPIO_71	SLIMBUS_DATA	V

GPIO	Net Name	Wakeup
GPIO_72	CODEC_RST_N	V
GPIO_73	RFFE2_CLK	
GPIO_74	RFFE2_DATA	
GPIO_75		
GPIO_76		
GPIO_77		
GPIO_78		
GPIO_79	B28_A_B_SEL	
GPIO_80	SW_PRX_MB1_B1B2_1	
GPIO_81	SW_PRX_MB1_B1B2_2	
GPIO_82	DRX_B41_B7_SEL	
GPIO_83		
GPIO_84		
GPIO_85	RF_CONN_DET_1	
GPIO_86	RF_CONN_DET_2	
GPIO_87	DRX_B1_2_34_SEL1	
GPIO_88	DRX_B1_2_34_SEL2	
GPIO_89		
GPIO_90	ANT_M1	
GPIO_91	ANT_M2	
GPIO_92	ANT_M3	
GPIO_93		
GPIO_94		
GPIO_95	WTR0_GSM_TX_PHASE_D0	
GPIO_96	WTR0_GSM_TX_PHASE_D1	
GPIO_97		
GPIO_98		
GPIO_99	RFFE1_CLK	
GPIO_100	RFFE1_DATA	
GPIO_101	CH0_SSBI_RX	
GPIO_102	CH0_SSBI_TX_GPS	
GPIO_103		
GPIO_104	TOP_AMP_BOOST_BP	
GPIO_105	GPS_EXT_LNA_EN	
GPIO_106	KB_VOL_UP	V
GPIO_107	KB_VOL_DOWN	V

GPIO	Net Name	Wakeup
GPIO_108		V
GPIO_109	HALL_SENSOR_INT	V
GPIO_110		V
GPIO_111		V
GPIO_112	TOP_AMP_BOOST_EN	
GPIO_113	FORCE_USB_BOOT	V
GPIO_114	FCAM_AVDD_EN	
GPIO_115		V
GPIO_116		
GPIO_117		V
GPIO_118		
GPIO_119		V
GPIO_120		

Function	IC	I2C Address
Gyro	BMG160	0x68
G-Sensor	BMC150	0x11
E-Compass	BMC150	0x13
PS+ALS	STK3310	0x48
Backlight	LM3630ATMX	0x38
TP	S2716	0x20
Flash LED Driver	LM3642TLX NOPB	0x63
Rear Cam.	IMX179QGH5-C	0x10
Front Cam.	IMX132TQH5-C	0x36
LCM Bias	TPS65132B0YFFR	0x3E

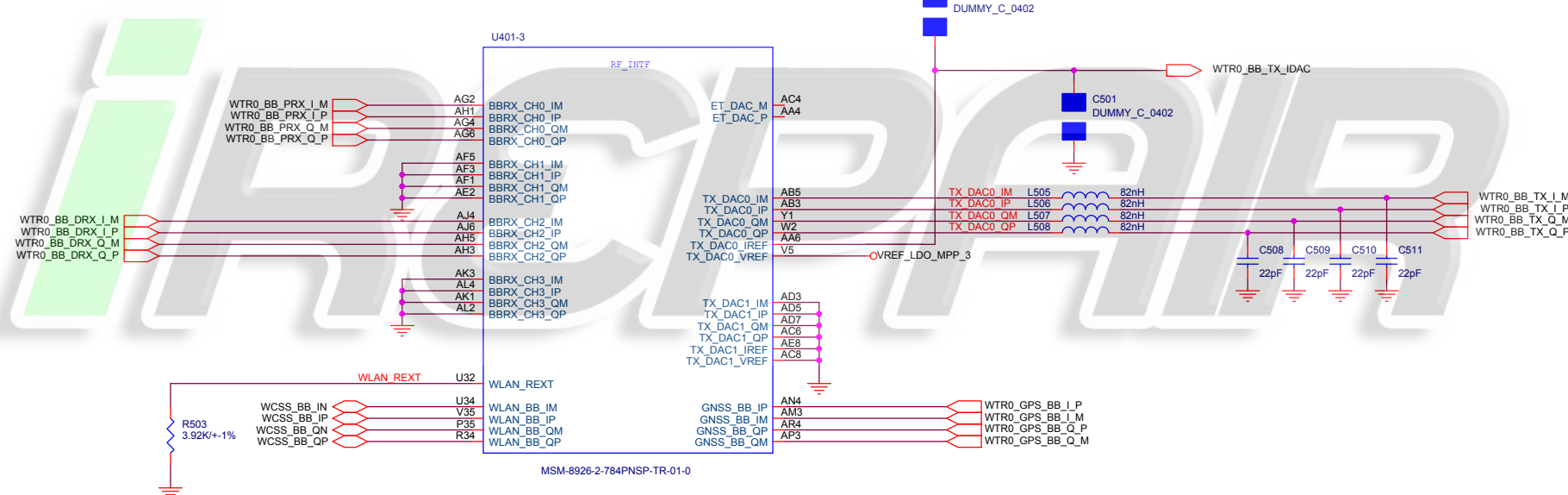
FIH Confidential and Proprietary			
Title	GPIO/I2C LIST		
Size A3	Document Number	Titan LTE	
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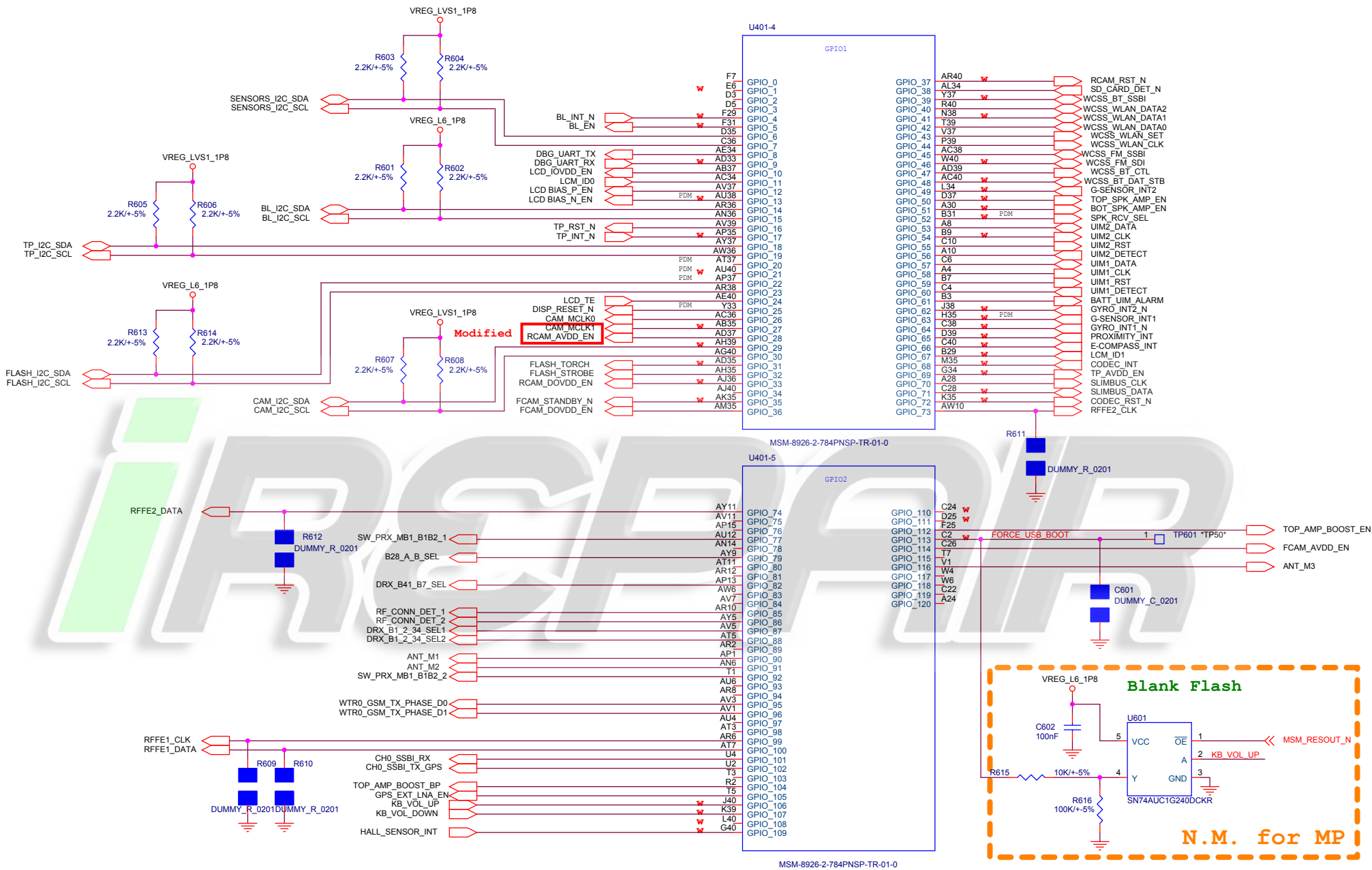
REAR CAMERA

FRONT CAMERA

DISPLAY



I/Q Usage Scenarios	ET	VDD A2 (pin AD9)	TX DAC1 I/Q	ETDAC P/M	TX DAC1 IREF	TX DAC1 VREF
TX DAC1 I/Q Unused	ET Unused	GND	GND	Floating	GND	GND



GPIOs	Corresponding fuse	Function
GPIO[82]	FAST_BOOT[1]	The fast boot options configure the external boot device used to boot from, as shown in the table below.
GPIO[83]	FAST_BOOT[2]	
GPIO[84]	FAST_BOOT[3]	Development board - BOOT_CONFIG general-purpose input/output (GPIOs) should be used.
GPIO[85]	FAST_BOOT[4]	Production board - FAST_BOOT fuses should be blown.

BOOT_CONFIG [4:1]	MAPPING
0b00000	eMMC (default) at SDC1, followed by USB from SDC2
0b00001	SDC2 followed by SDC1
0b00010	eMMC at SDC1
0b00011	USB

MSM Boot Config Pin: Internal PD

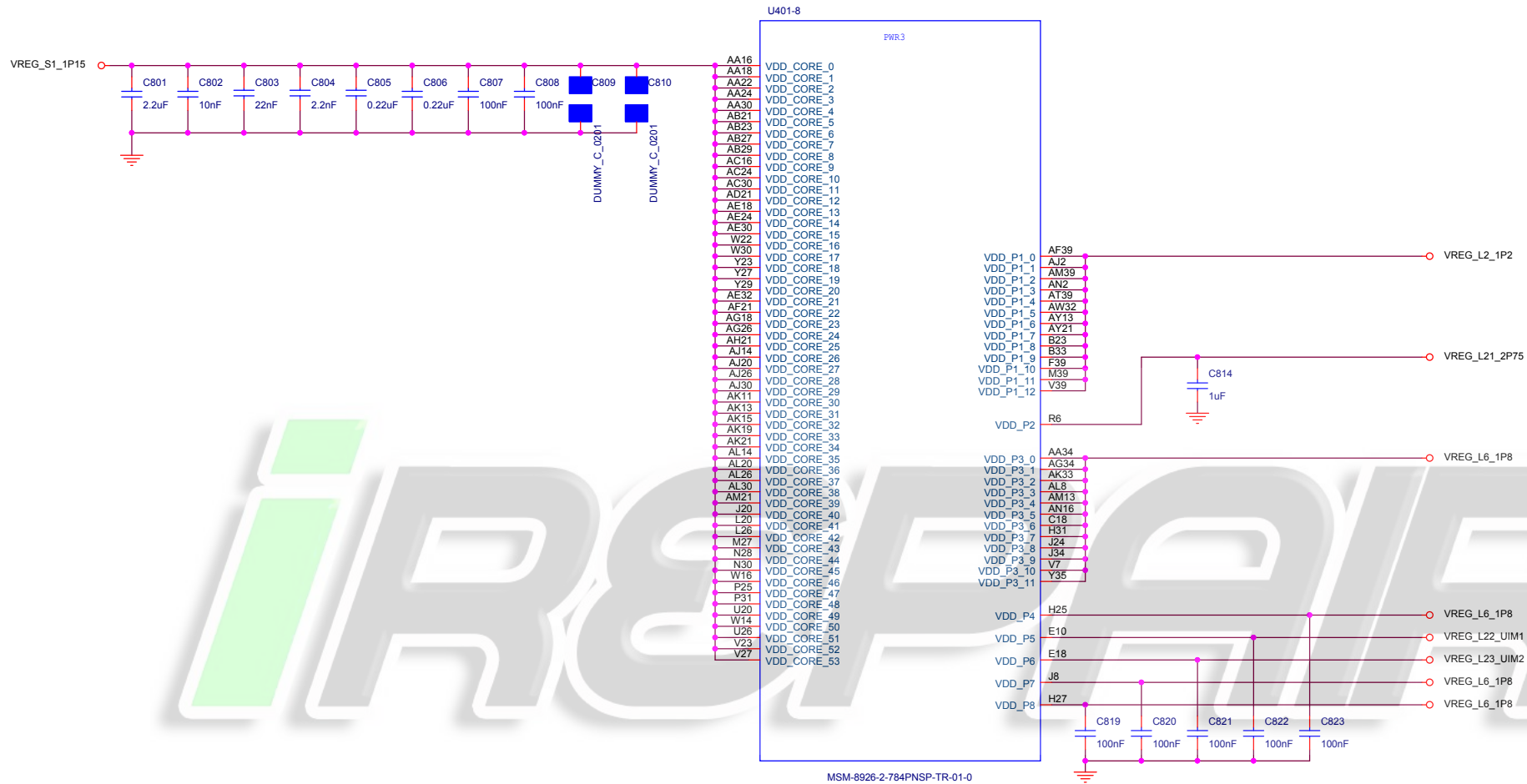
FORCE_USB_BOOT (GPIO_113) is checked first during the boot device detection prior to BOOT_CONFIG GPIOs.

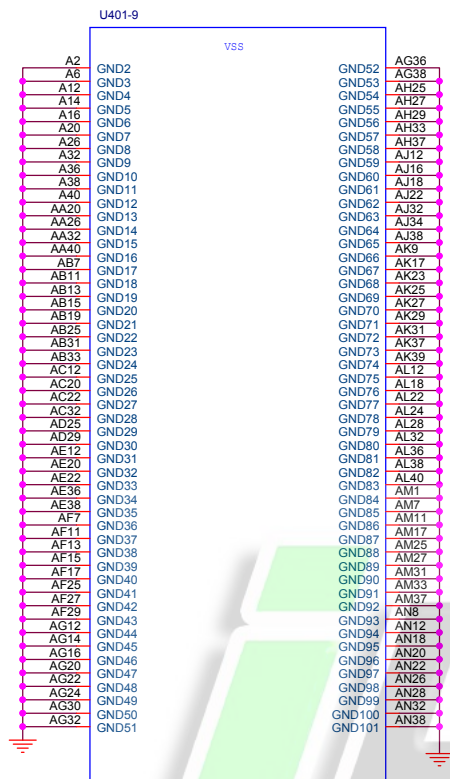
GPIO_113 = 1 will force the MSM device to boot from USB_HS port.

The fuse FORCE_USB_BOOT_DISABLE can be blown to disable the feature to force USB boot using GPIO_113.

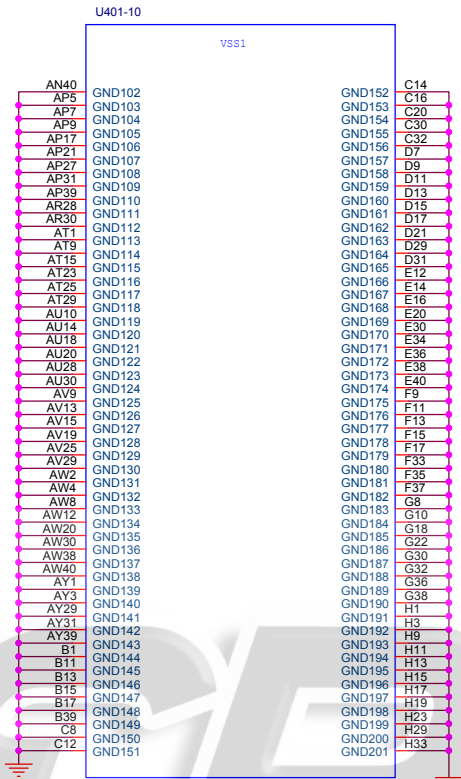
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Title	MSM8926 GPIO		
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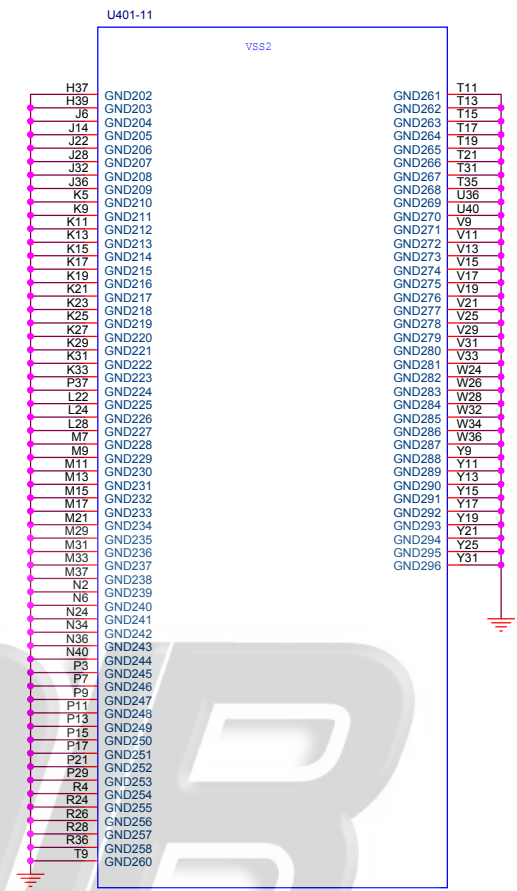




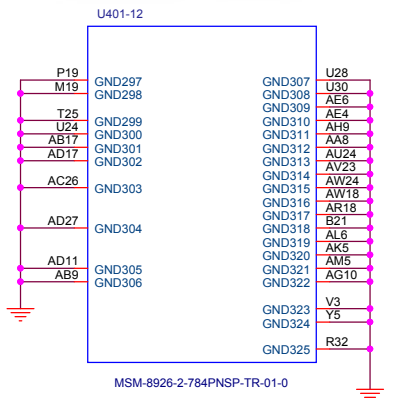
MSM-8926-2-784PNSP-TR-01-0



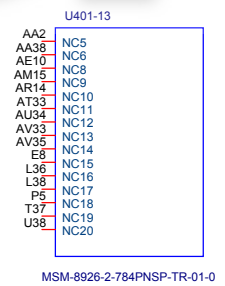
MSM-8926-2-784PNSP-TR-01-0



MSM-8926-2-784PNSP-TR-01-0

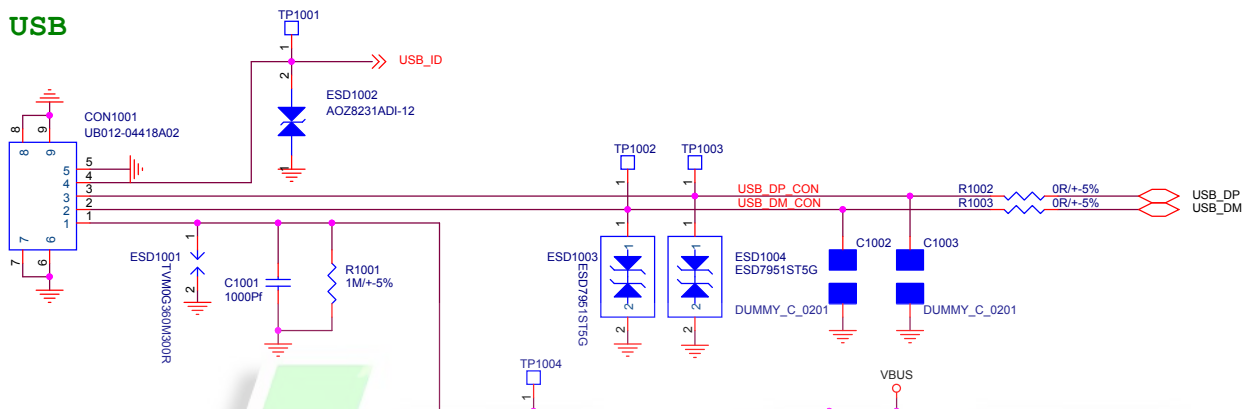


MSM-8926-2-784PNSP-TR-01-0

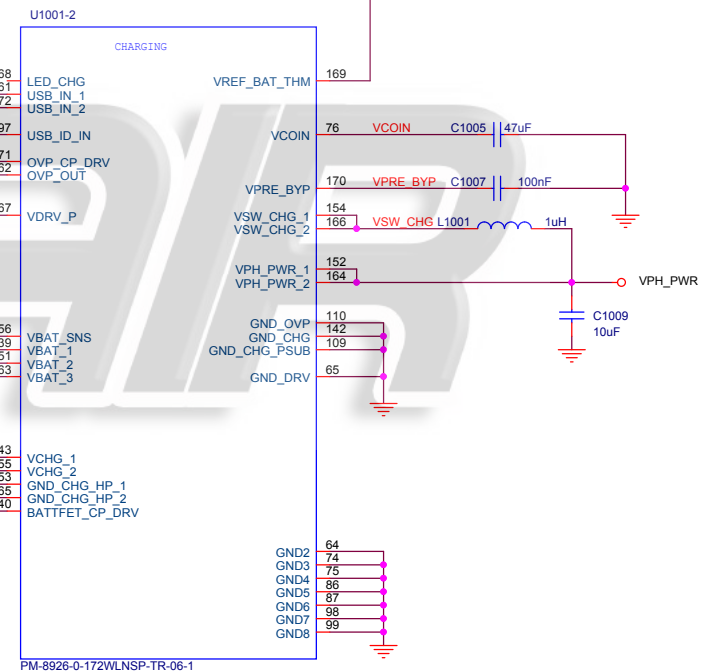
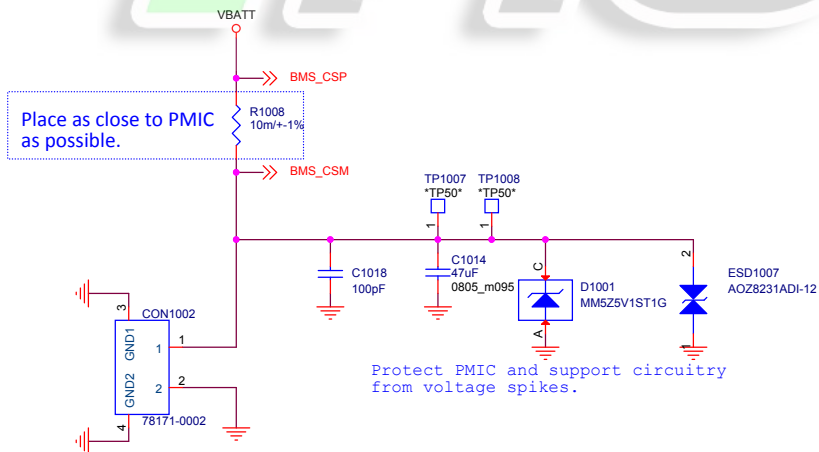


MSM-8926-2-784PNSP-TR-01-0

USB

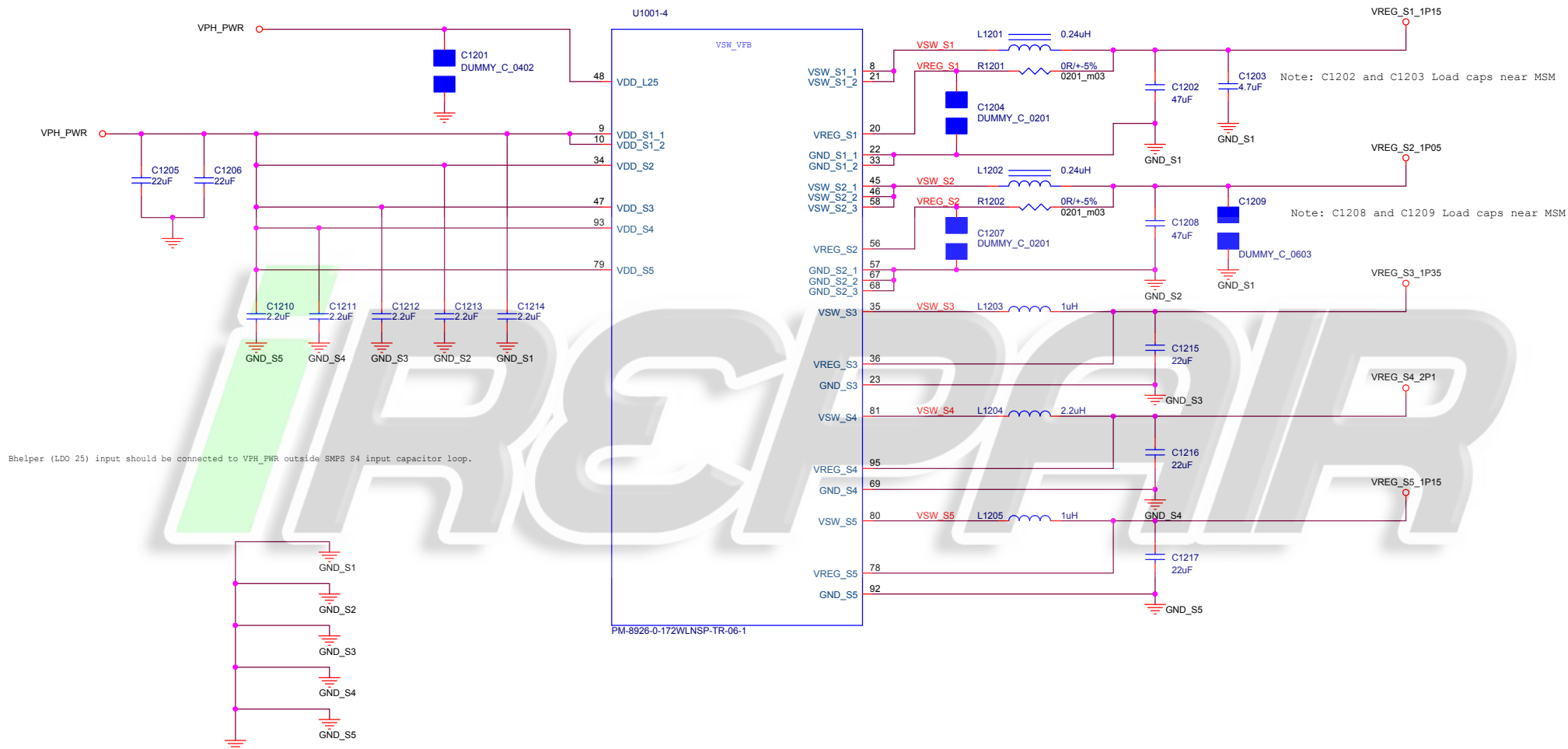


BATTERY CONNECOT



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Title	PM8926_CHARGING		
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Note:0.1R shunts are for proto only.

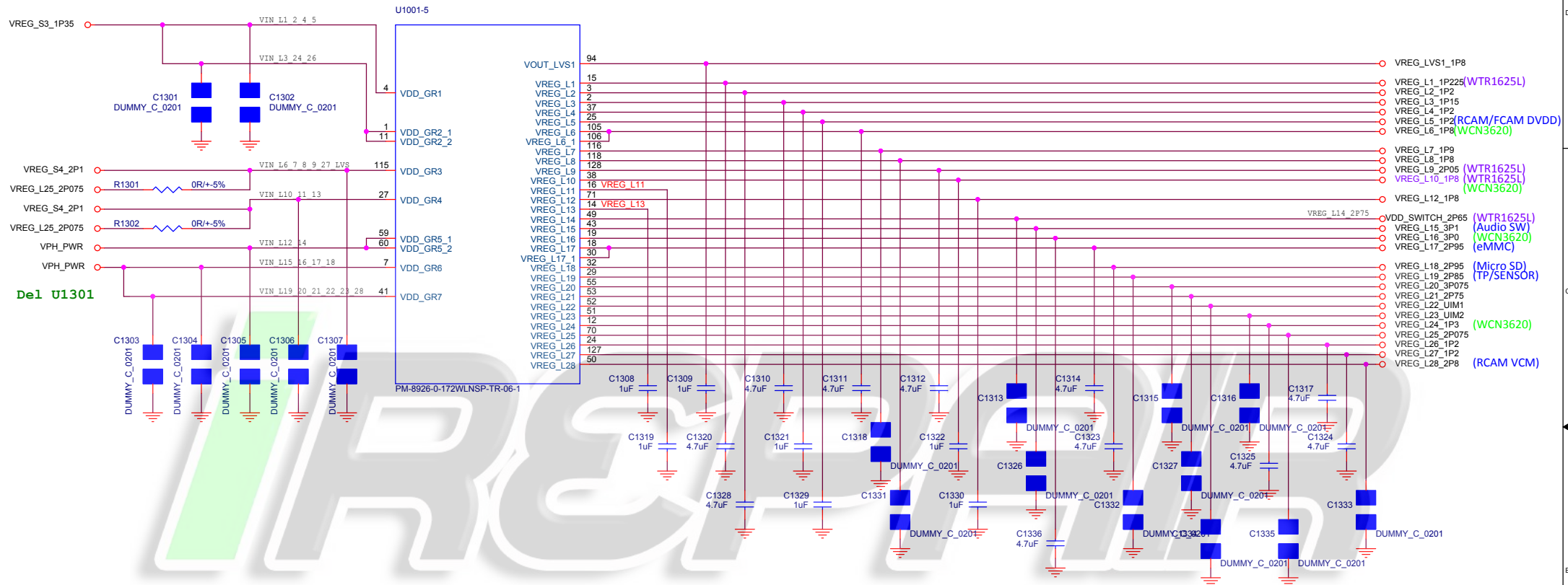
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Title	PM8926_SMPS		
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RF power notice

WTR1625L: VREG_L9_2P05 , VREG_L1_1P225, VREG_L10_1P8

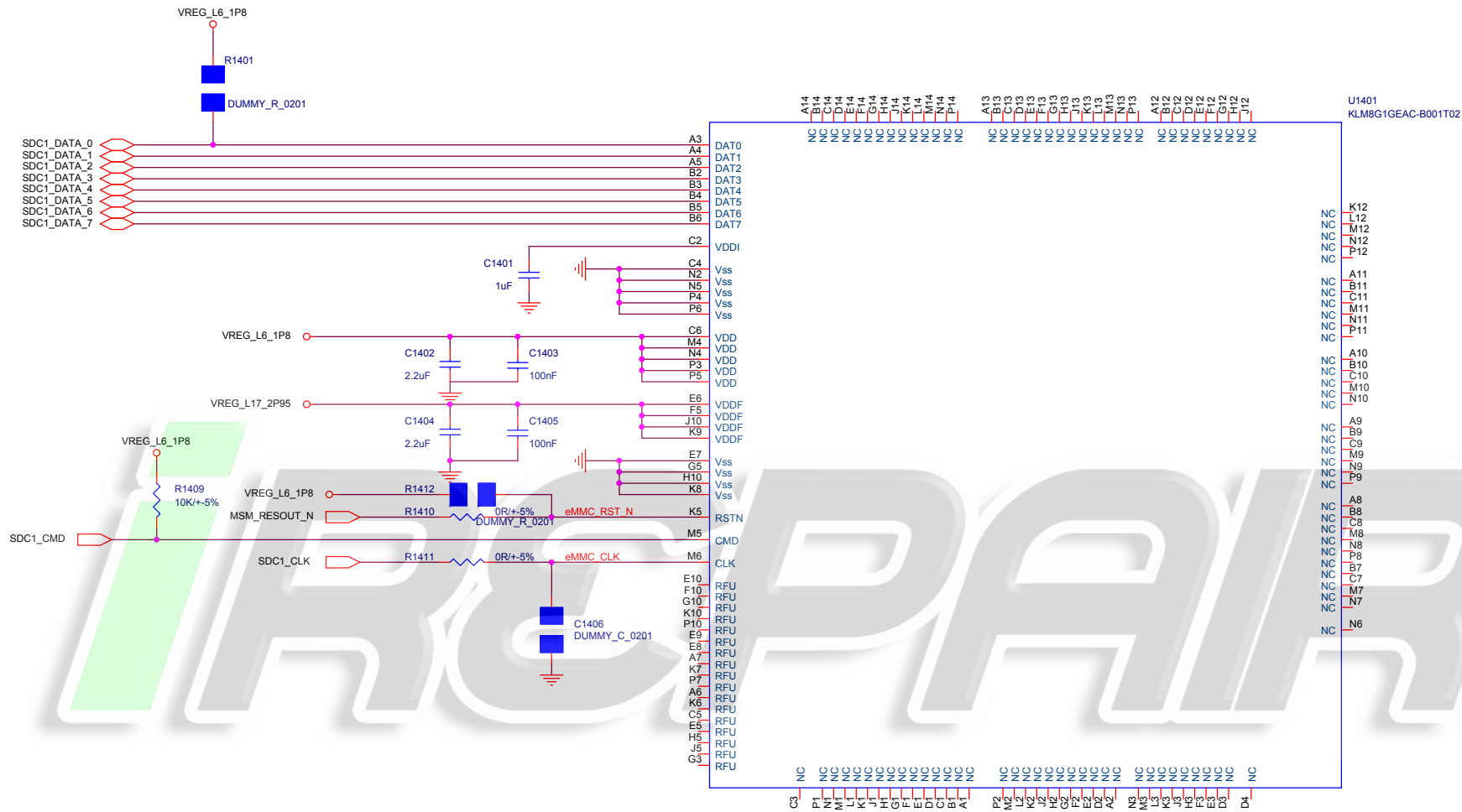
WCN3620: VREG_L6_1P8, VREG_L16_3P0, VREG_L24_1P3, VREG_L10_1P8(XO)

RF switch power: VDD_SWITCH_2P65(VREG_L14_2P75)



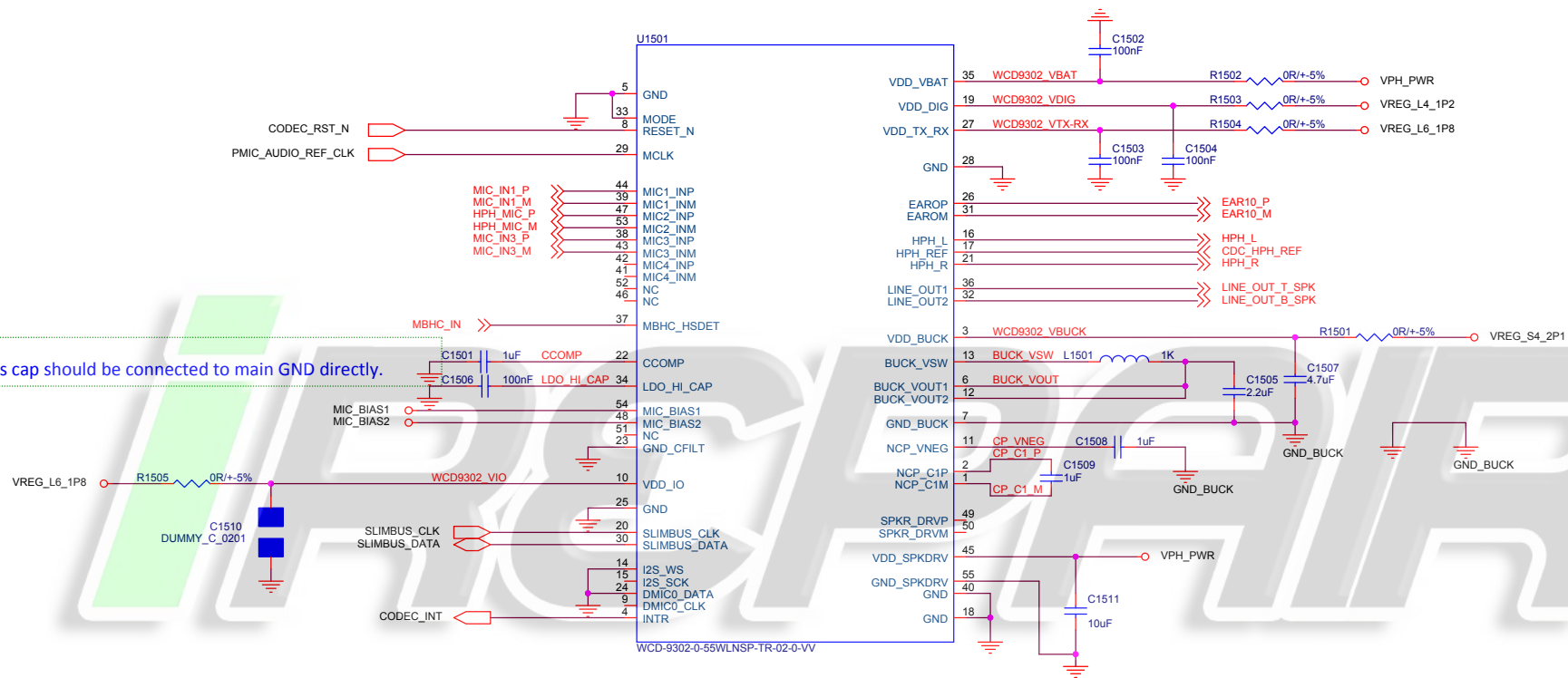
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Title		PM8926_LDO	
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U1402-POP
LPDDR2-8Gb EDB8132B3PB-1D-F (SA0B8132010)

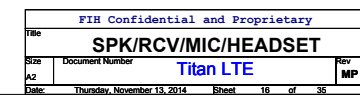
Note:
This bypass cap should be connected to main GND directly.

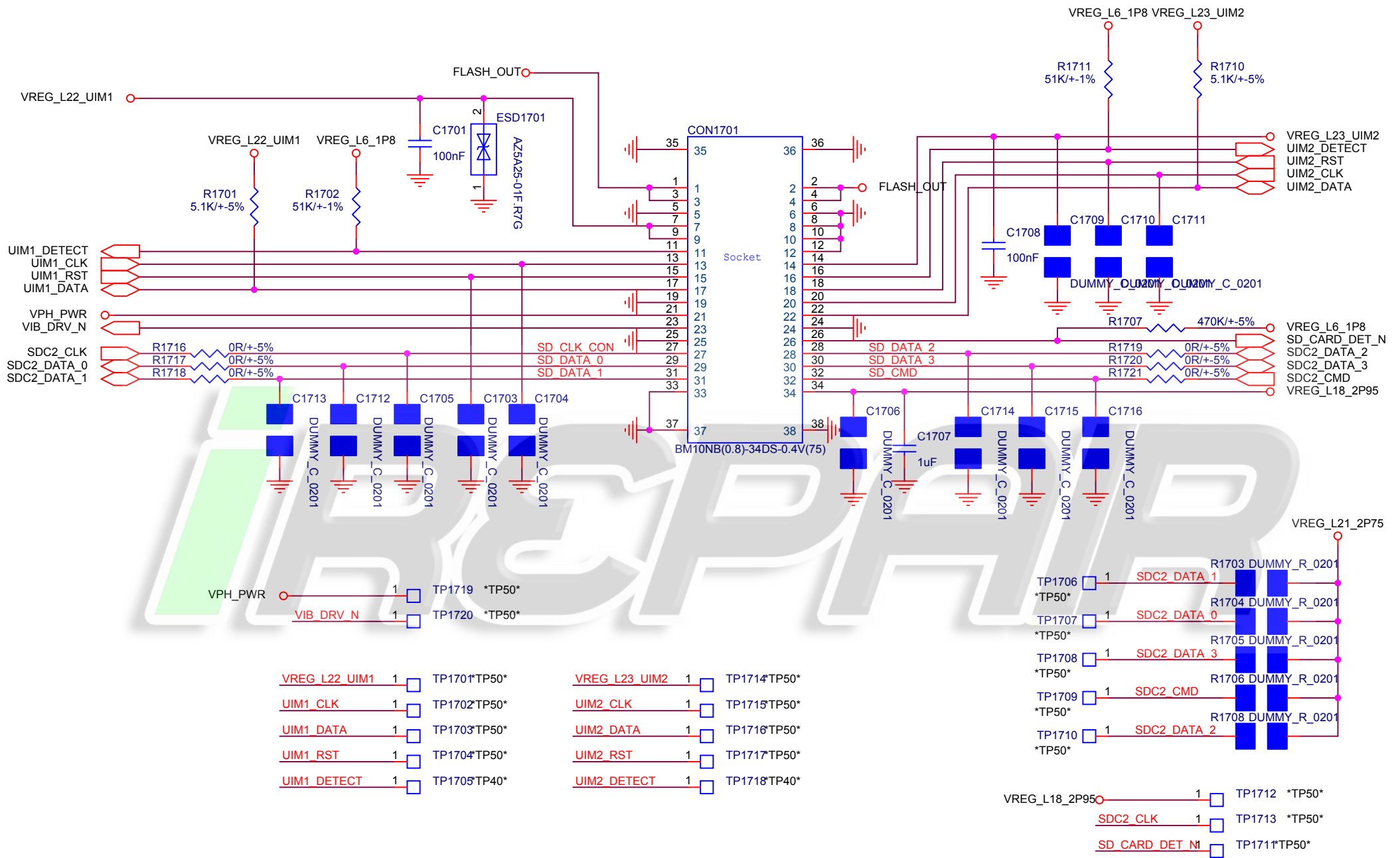


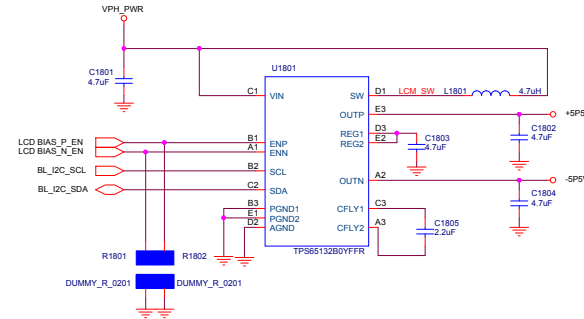
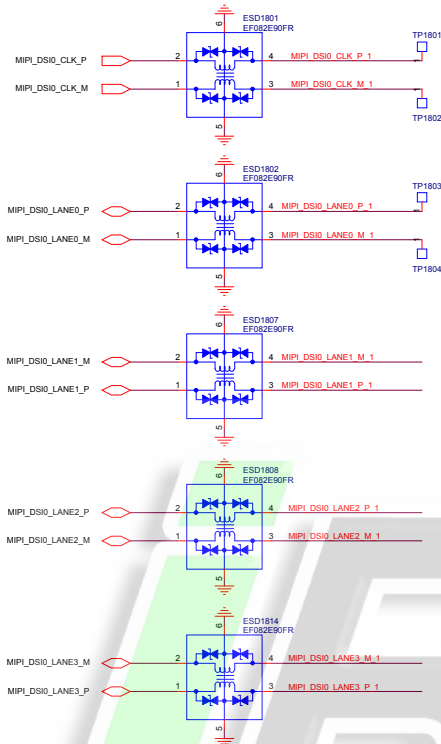
Note:
• Pin 18, 23 and pin 40 should be connected to main GND directly.
• Pin 55 should be connected to the GND side of C1511 and directly connected to the main GND with dedicated vias.

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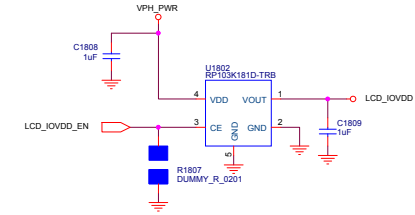
Title	WCD9302		
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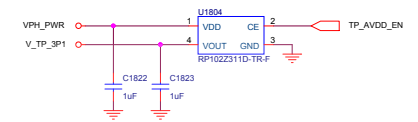




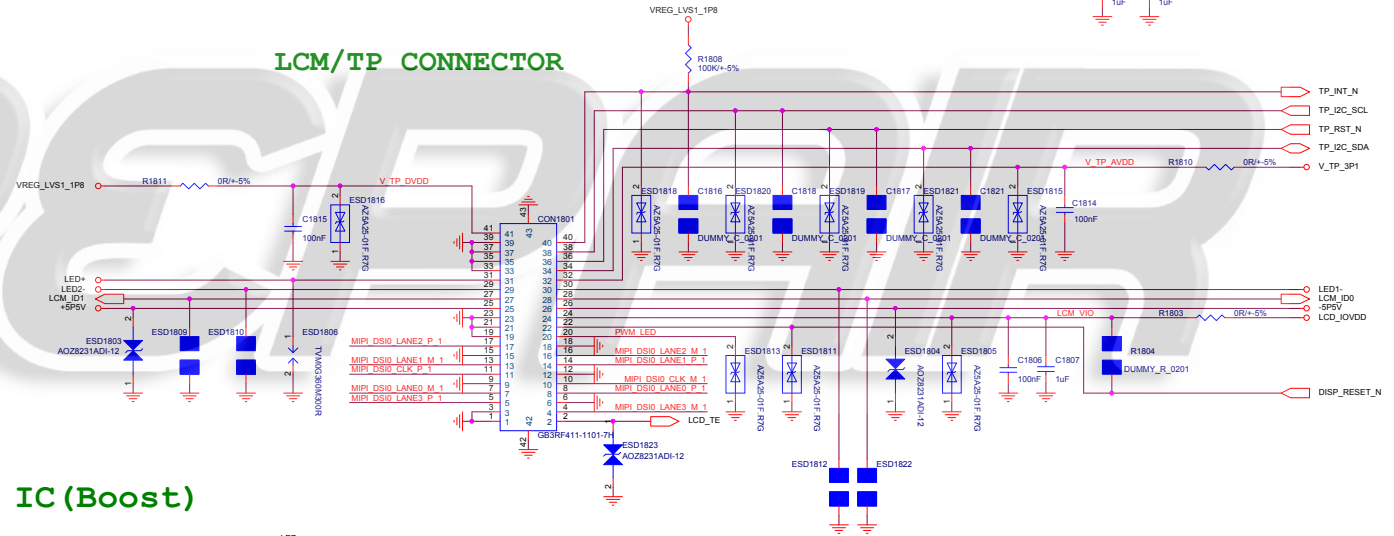
LCM IO POWER



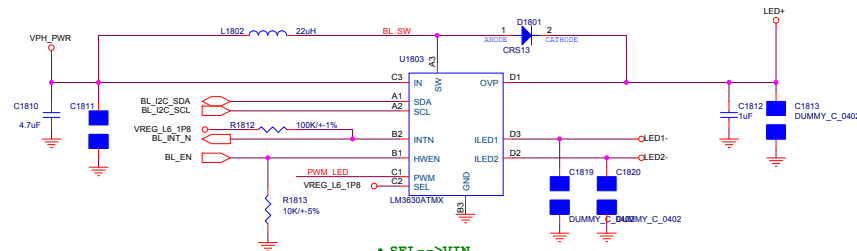
TP AVDD POWER



LCM/TP CONNECTOR

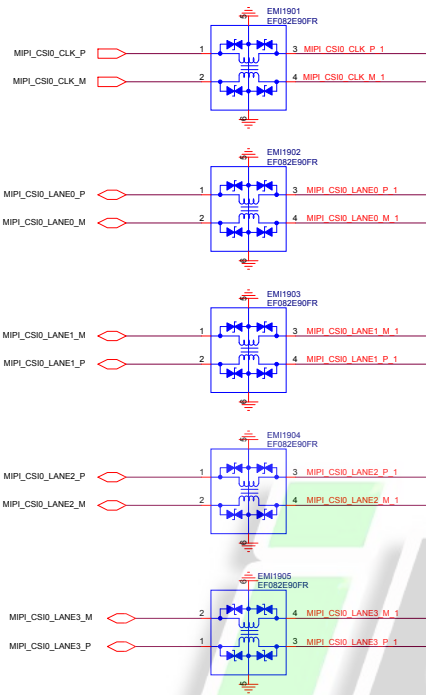


Back Light IC (Boost)

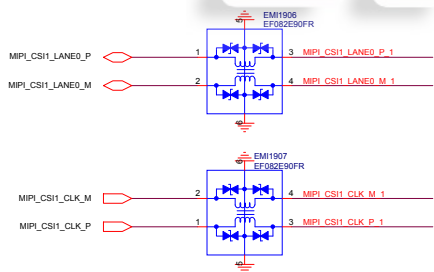


- SEL-->VIN
- I2C address: 0x38
- SEL-->GND
- I2C address: 0x36

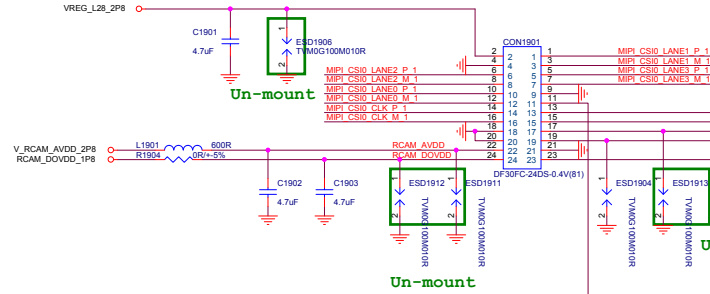
REAR CAMERA MIPI EMI



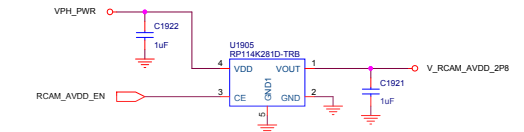
FRONT CAMERA MIPI EMI



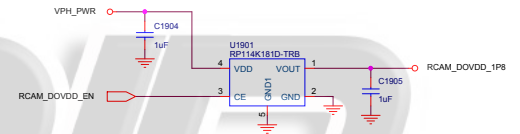
8M REAR CAMERA



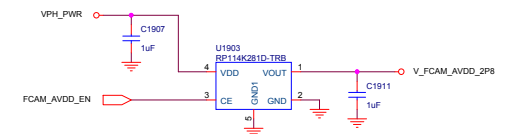
REAR CAM AVDD POWER



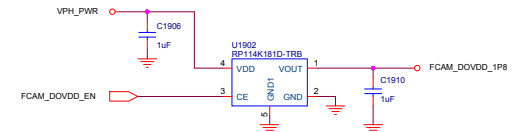
REAR CAM DOVDD POWER



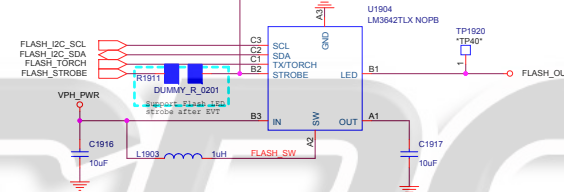
FRONT CAM AVDD POWER



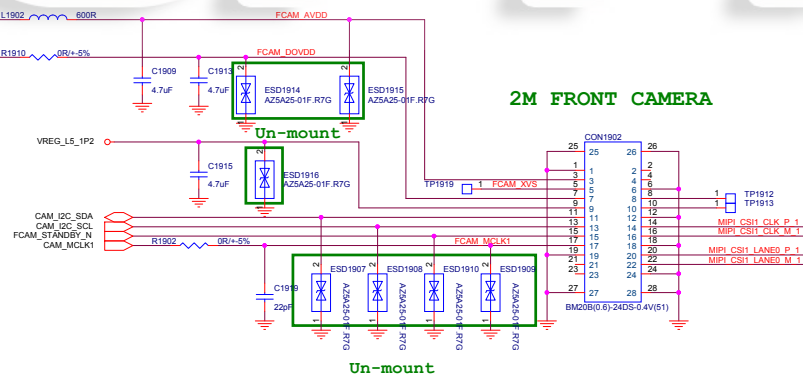
FRONT CAM DOVDD POWER

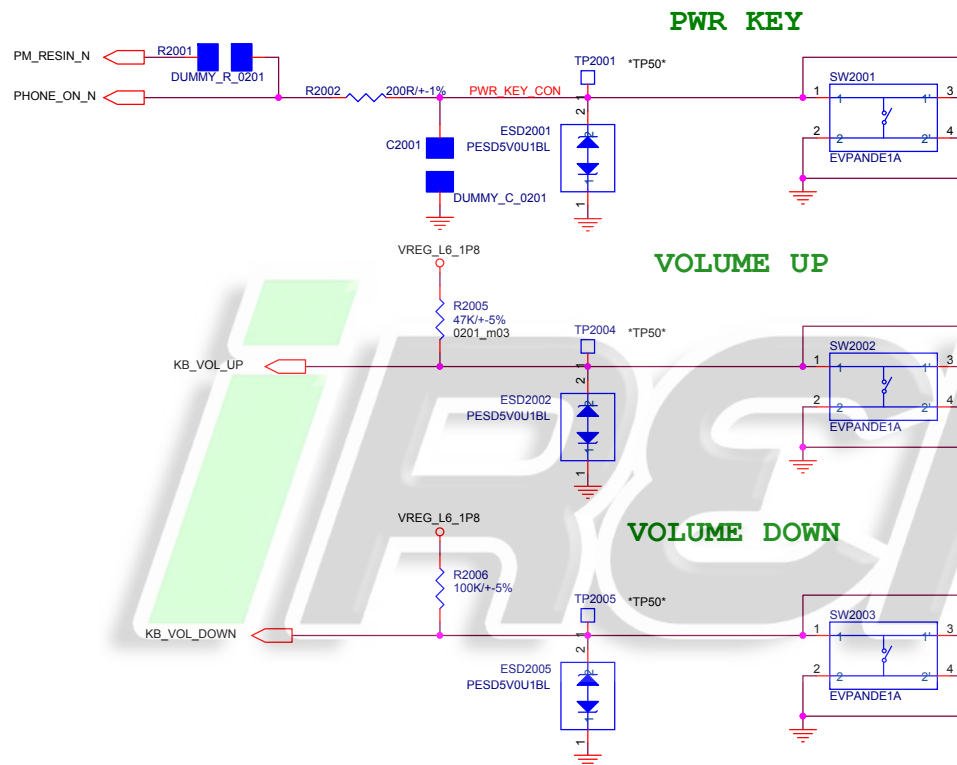


Flash LED Driver

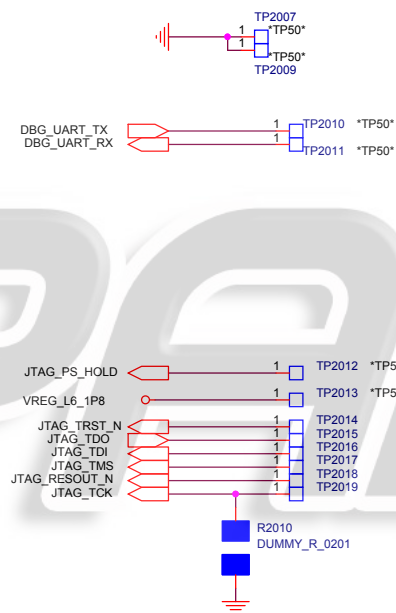


2M FRONT CAMERA

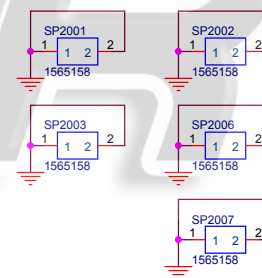




TEST POINTS



GND SPRING



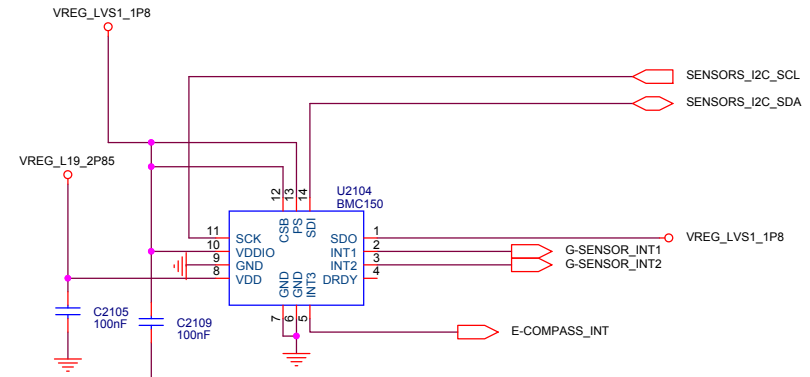
RF Cable Hold Housing



Del SP2008 on 0901

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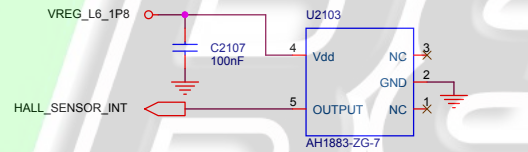
G-Sensor + E-Compass



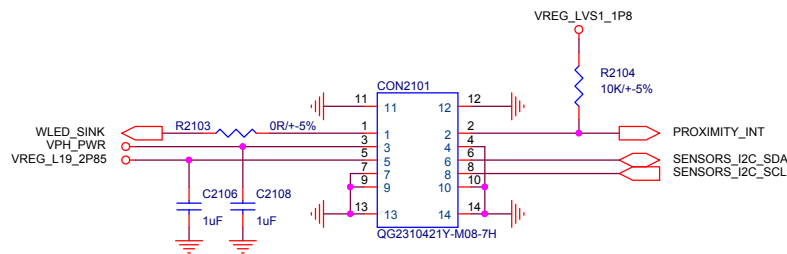
I2C address:

- SDO-->VIO
G-Sensor: 0x11 E-Compass: 0x13
- SDO-->GND
G-Sensor: 0x10 E-Compass: 0x12

HALL SENSOR

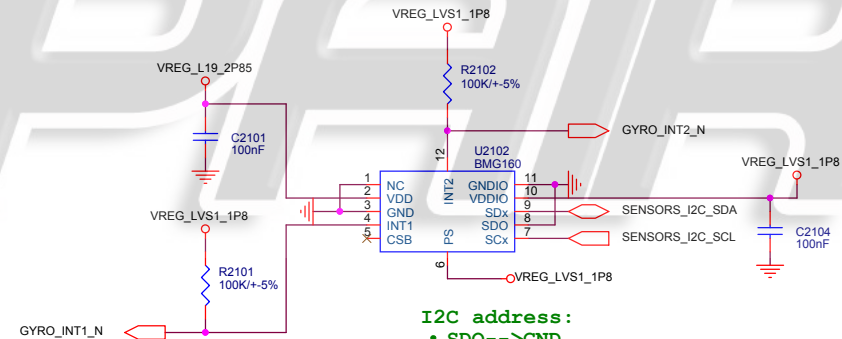


P+ALS SENSOR/INDICATOR CONNECTOR



I2C address: 0x90

Gyro



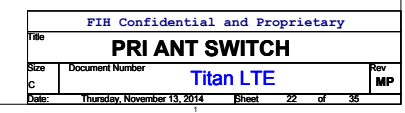
I2C address:

- SDO-->GND
Gyro: 0x68
- SDO-->VIO
Gyro: 0x69

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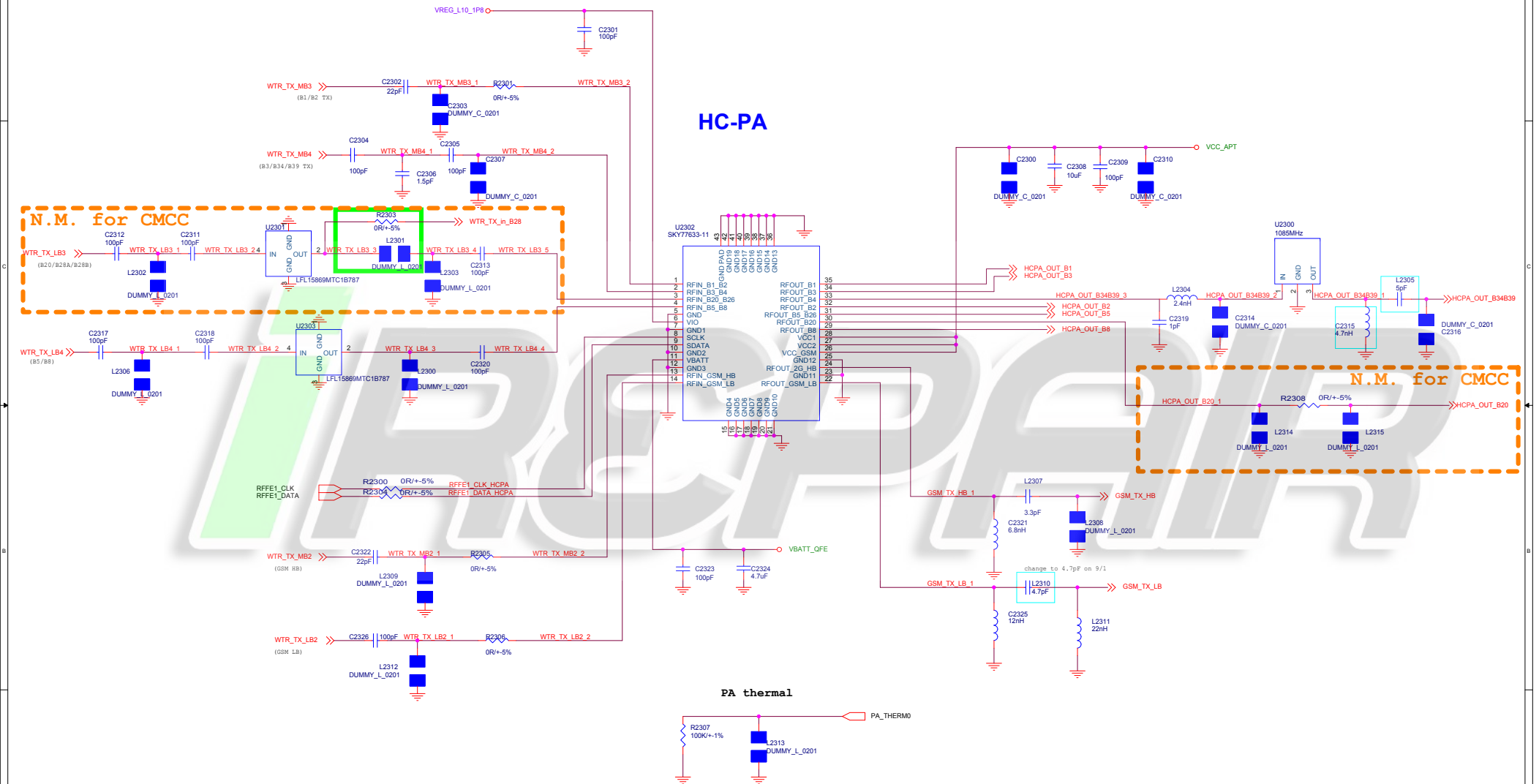
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LTE Main
sub-board



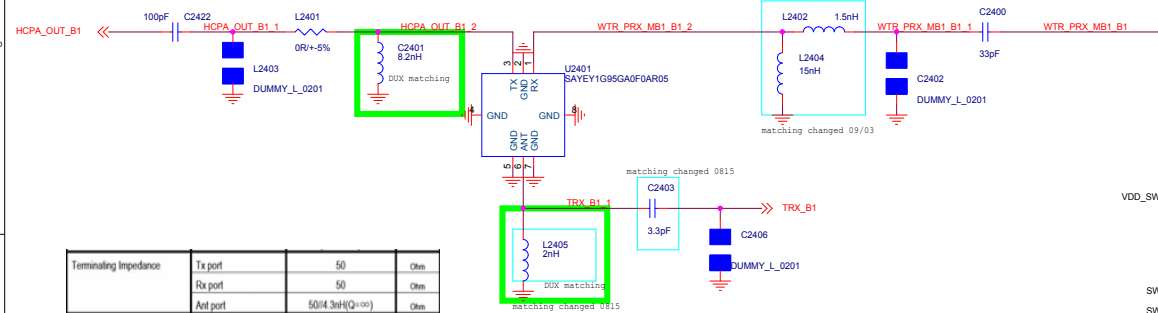
Filter No.	Pass band (MHz)	System
1	1930~1990	G1900-R
2	1805~1880	G1800-R

GSM Qual band, UMTS B1/B2/B3/B5/B8/B20, B34/B39



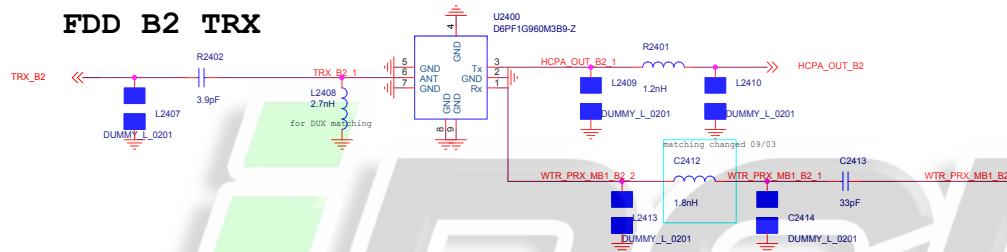
Duplexer B1/B2/B3/B34/B39

FDD B1 TRX

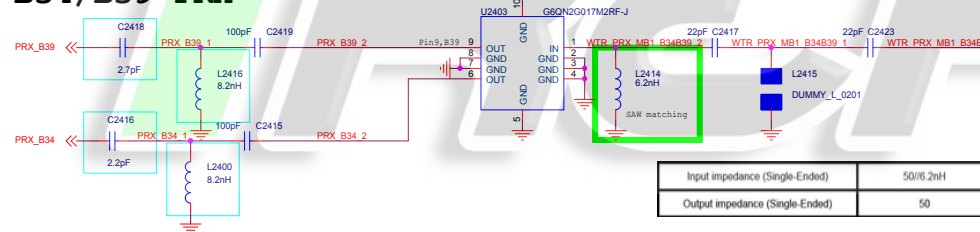


Terminating Impedance	Tx port	50	Ohm
	Rx port	50	Ohm
	Ant port	50/(4.3nH @ 100)	Ohm

FDD B2 TRX

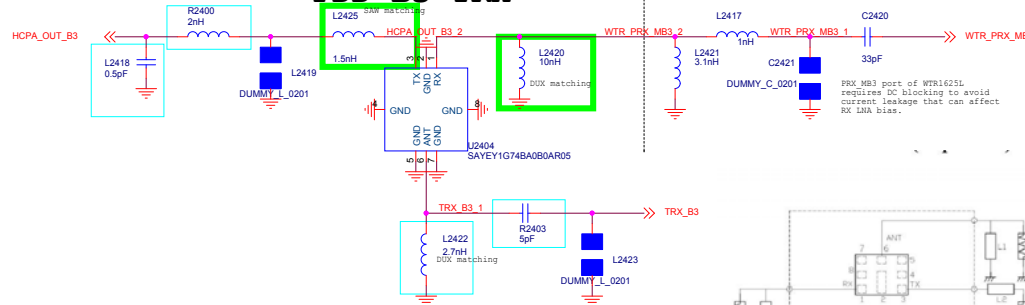


B34/B39 PRX



Input impedance (Single-Ended)	50/6.2nH
Output impedance (Single-Ended)	50

FDD B3 TRX

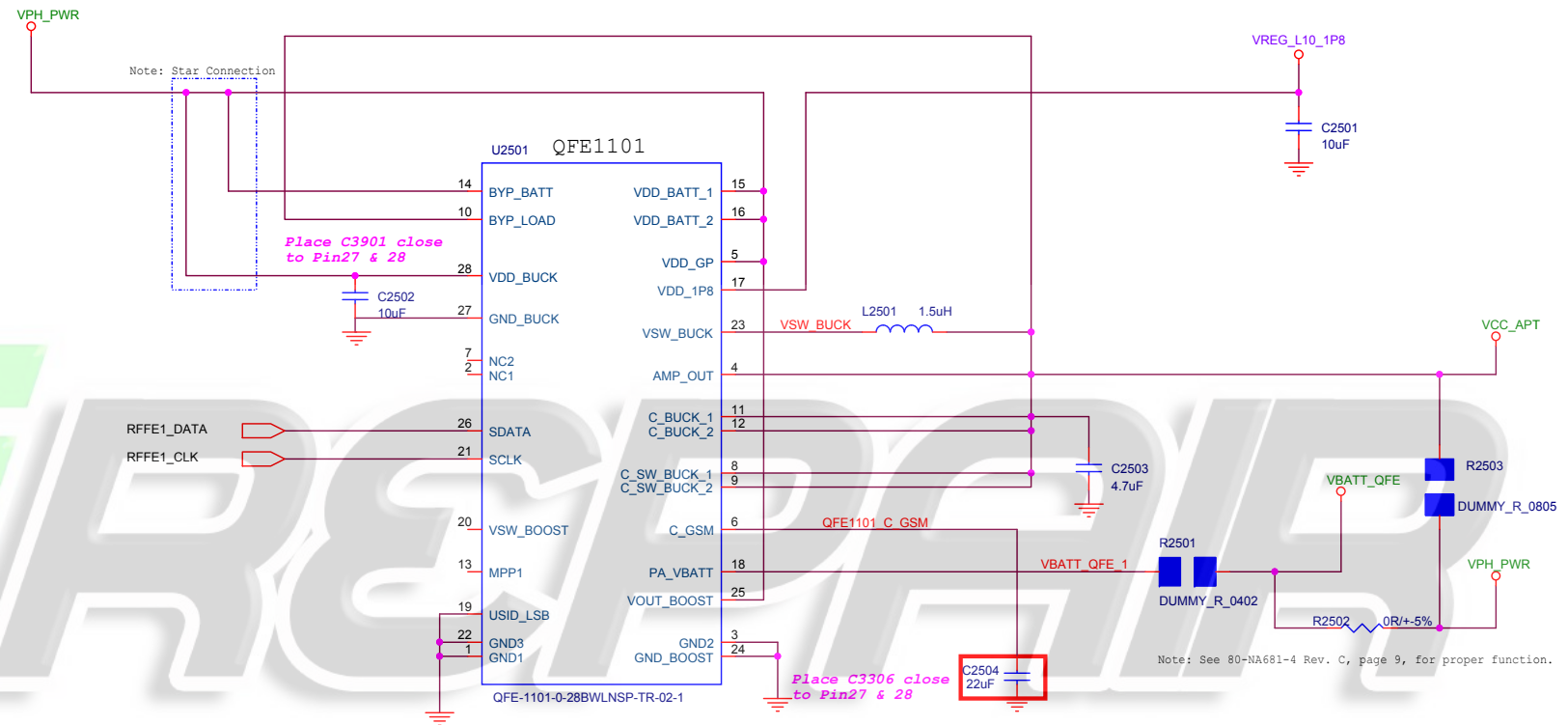


- R1 : 50 ohm
- R2 : 50 ohm
- R3 : 50 ohm
- L1 : 3.9 nH(Ideal inductor)
- L2 :series 1.5 nH(Ideal inductor)
- L3 : 10 nH(Ideal inductor)

QFE1101

80-NA437-46 Rev.B

Note: QFE1100 ES2 is available end of February 2013.
QFE1100 ES2 + L3304 + C3303 changes are needed to support
low voltage battery and LTE 1RB over all technology.



Note: QFE1100 ES2 pin 13 connection to PMIC HKADC is for ET envelope gain calibration.

Note: Pin 13 cannot be pulled up to VDD:
- If using PMIC8941, please config PMIC Internal Pull-up switch to "NC" via software setting.
- If using PMIC8019, please remove external Pull-up resistor on Baseband card.

Place C3306 close to Pin27 & 28

Note: Replace with 33uF. Place second cap on back of the PWB

Note: See 80-NA681-4 Rev. C, page 9, for proper function.

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			Rev	MP

WTR1625 (L)

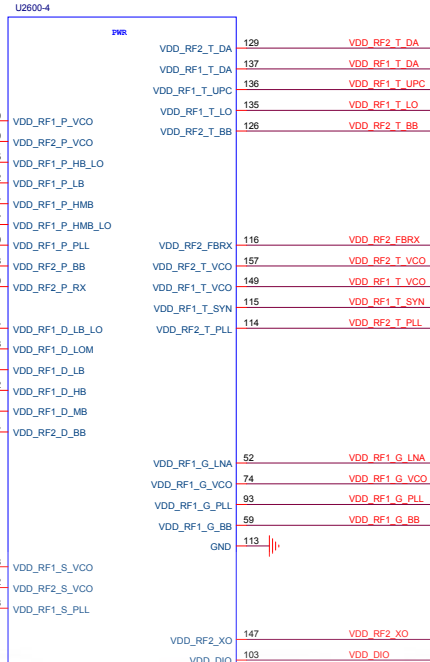
PWR

DISTRIBUTION

VDD_RF1_P_VCO 90
VDD_RF2_P_VCO 80
VDD_RF1_P_HB_LO 25
VDD_RF1_P_LB 72
VDD_RF1_P_HMB 34
VDD_RF1_P_HMB_LO 57
VDD_RF1_P_PLL 79
VDD_RF2_P_BB 98
VDD_RF2_P_PRX 100

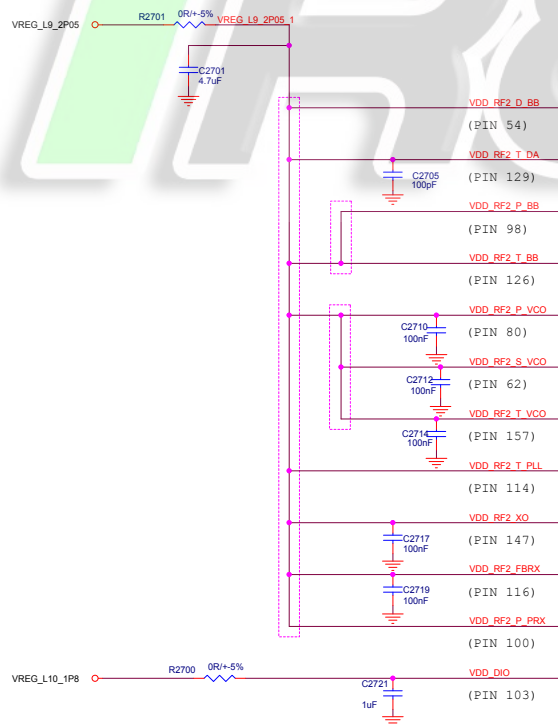
VDD_RF1_D_LB_LO 14
VDD_RF1_D_LOM 38
VDD_RF1_D_LB 31
VDD_RF1_D_HB 22
VDD_RF1_D_MB 11
VDD_RF2_D_BB 54

VDD_RF1_S_VCO 48
VDD_RF2_S_VCO 62
VDD_RF1_S_PLL 78

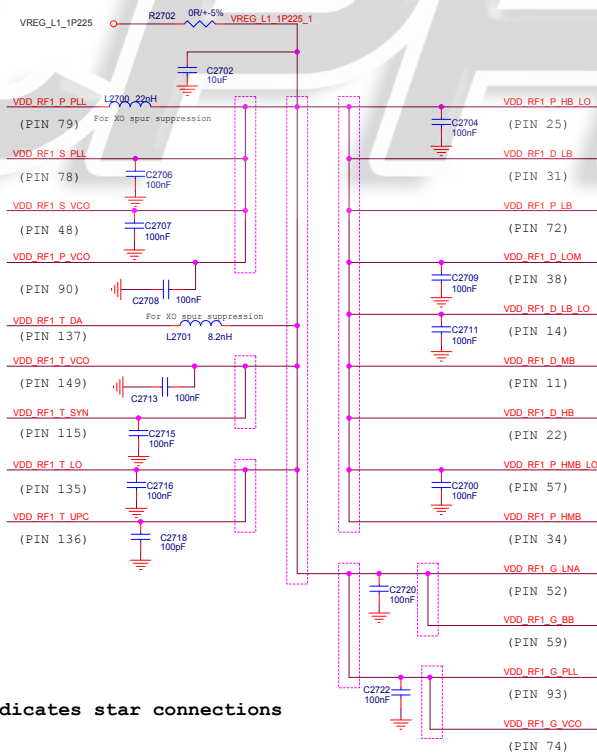


WTR-1625L-0-164WLN5P-TR-00-0

WTR POWER DISTRIBUTION 1.8V & 2V

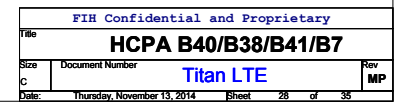


WTR POWER DISTRIBUTION 1.225V



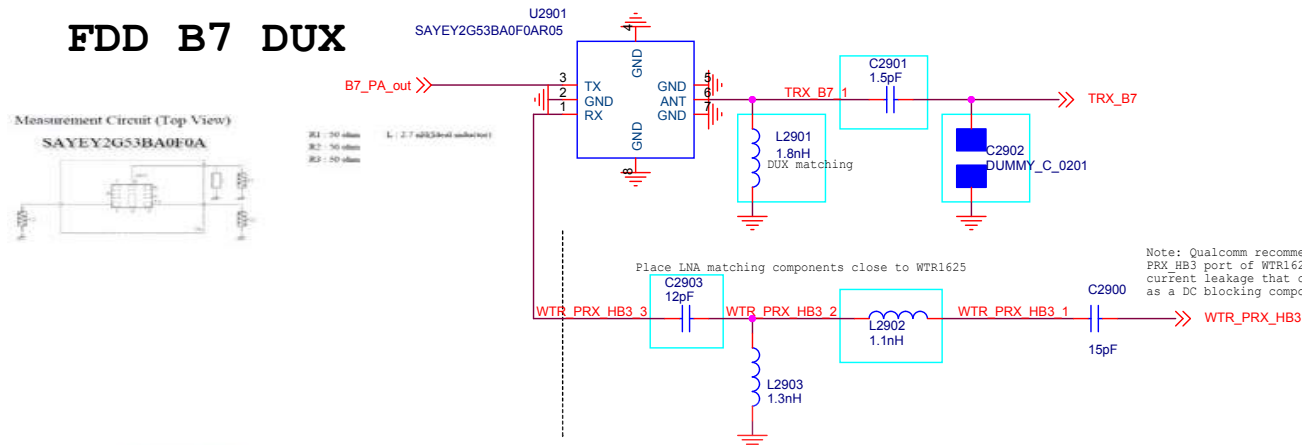
Indicates star connections

N.M. for CMCC

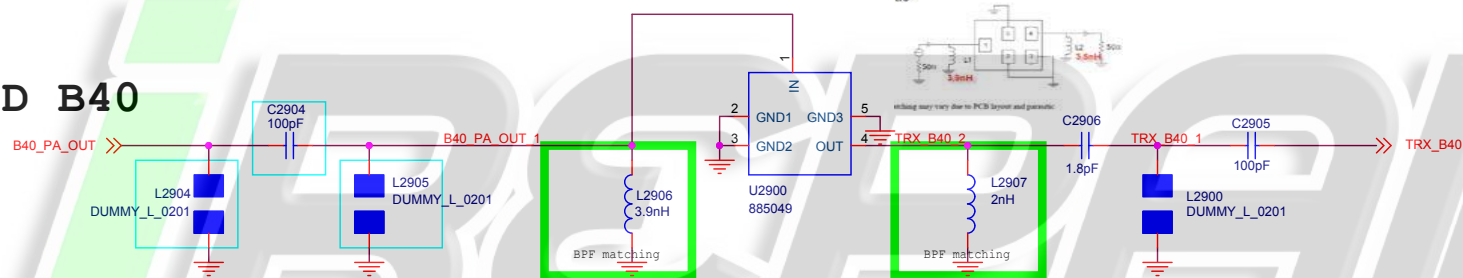


DUX B40/B38/B41/B7

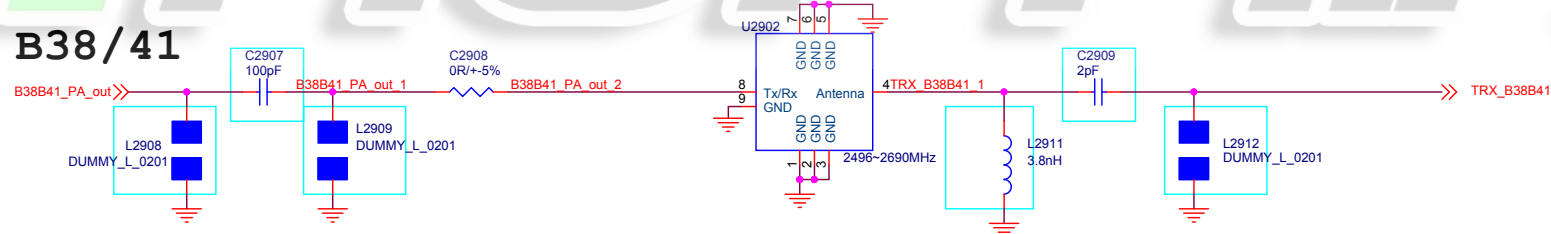
FDD B7 DUX



TDD B40



TDD B38/41



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Title DUX B40/B38/B41/B7

Size Document Number

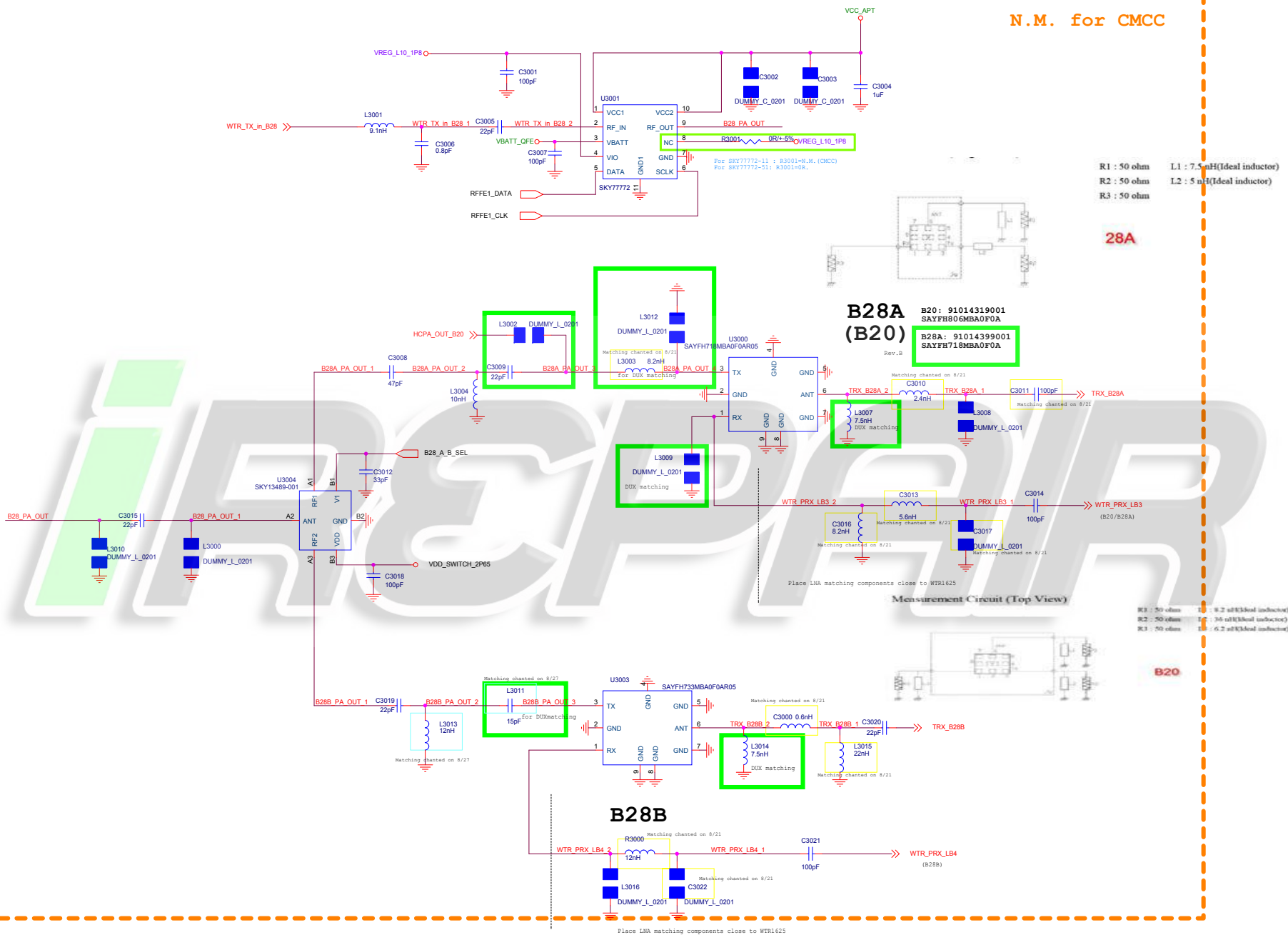
Titan LTE

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FDD B28 (B20) PA&TRX

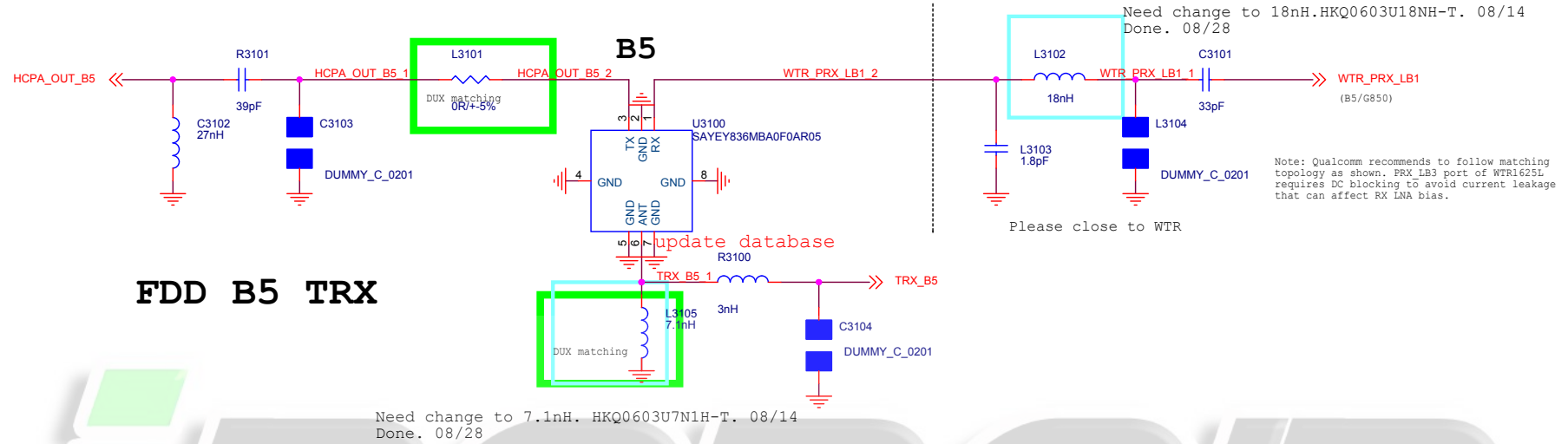
N.M. for CMCC



DUX B5/B8

R1 : 50 ohm L1 : 7.2 nH(Ideal inductor)
R2 : 50 ohm L2 : 25 nH(Ideal inductor)
R3 : 50 ohm

B5



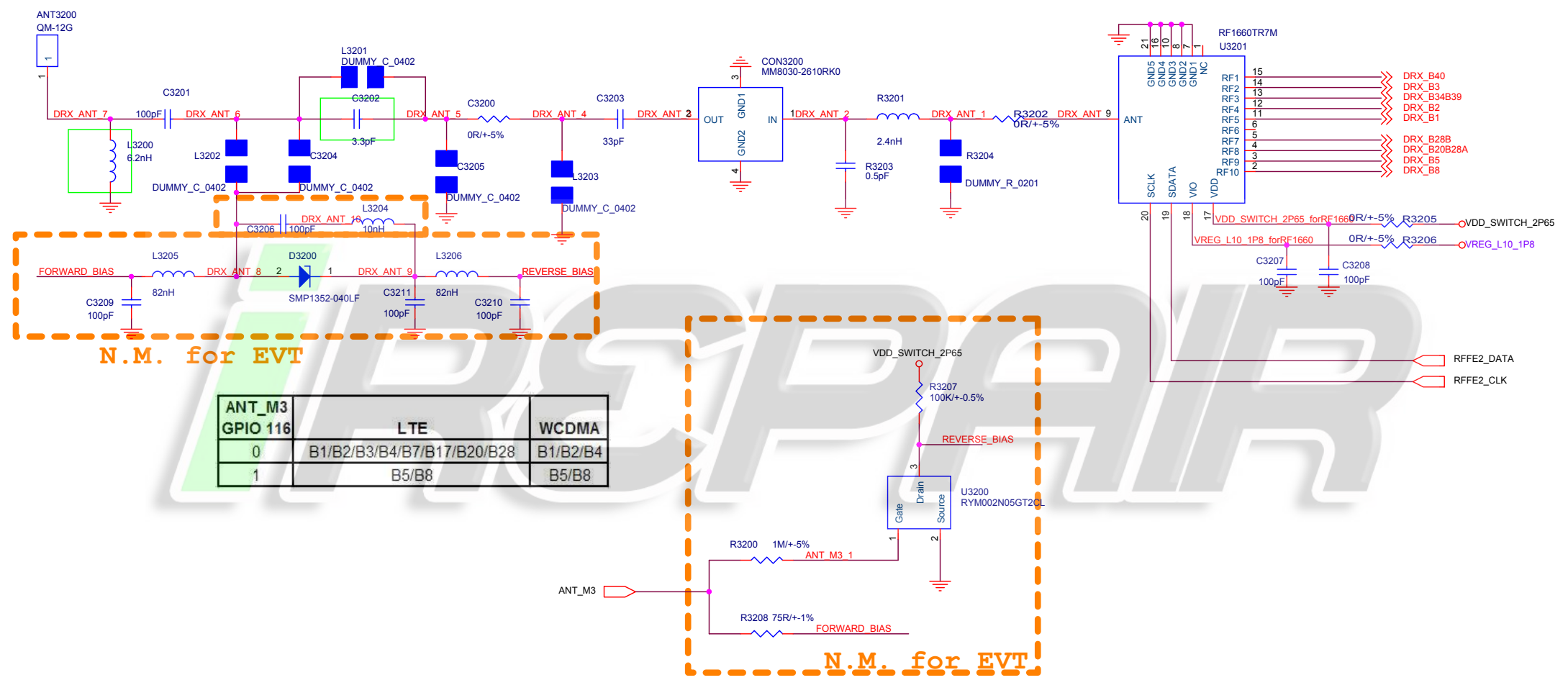
FDD B5 TRX

FDD B8 TRX

Terminating Impedance	Tx port	50
	Rx port	50
	Ant port	50 // 7.5nH (Q=∞)

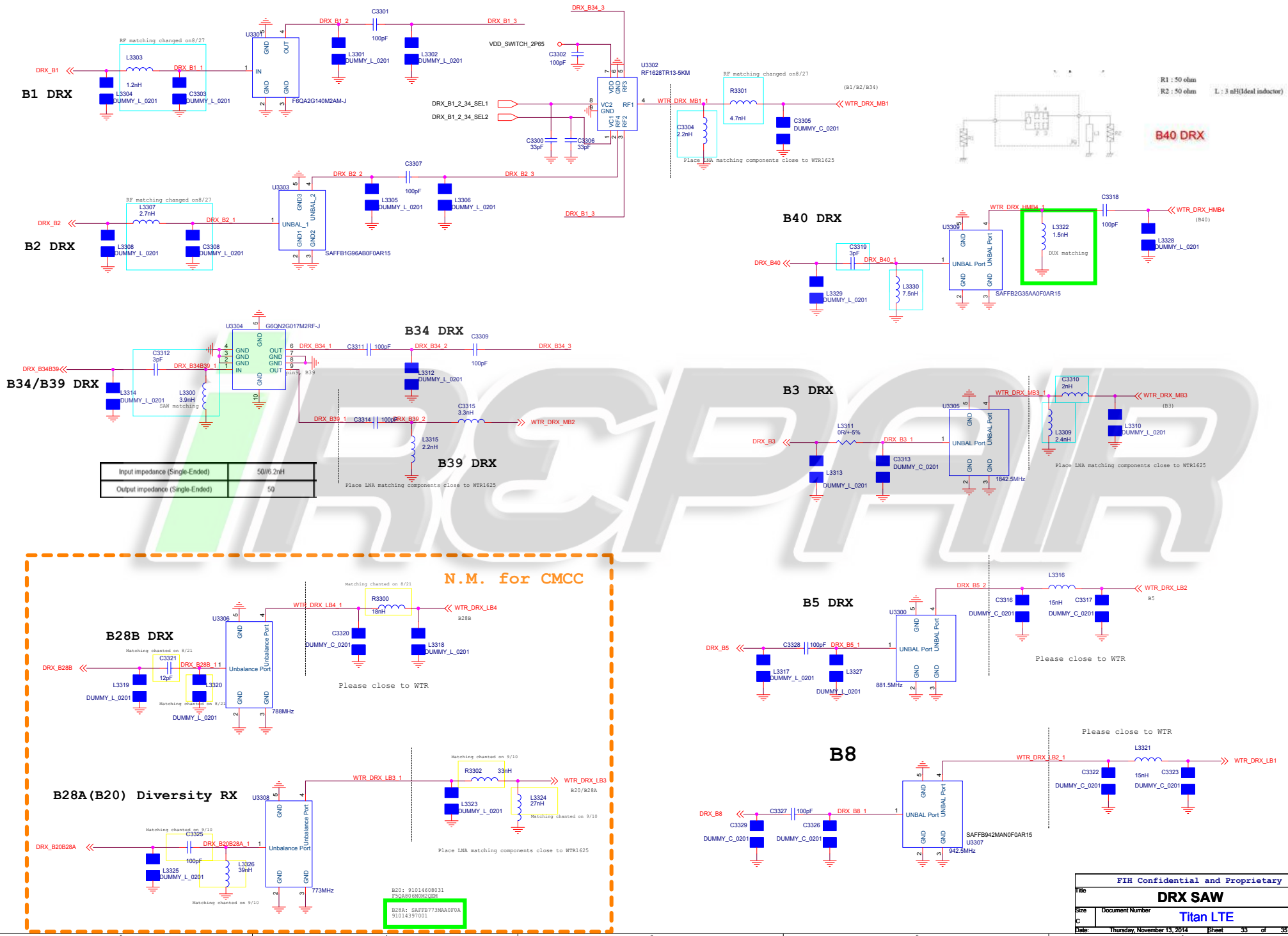
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DRX FEM

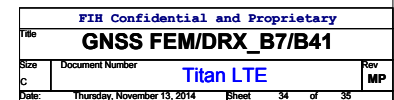


ANT_M3 GPIO 116	LTE	WCDMA
0	B1/B2/B3/B4/B7/B17/B20/B28	B1/B2/B4
1	B5/B8	B5/B8

DRX SAW



Place LNA matching components close to WTR1625



WCN3620

